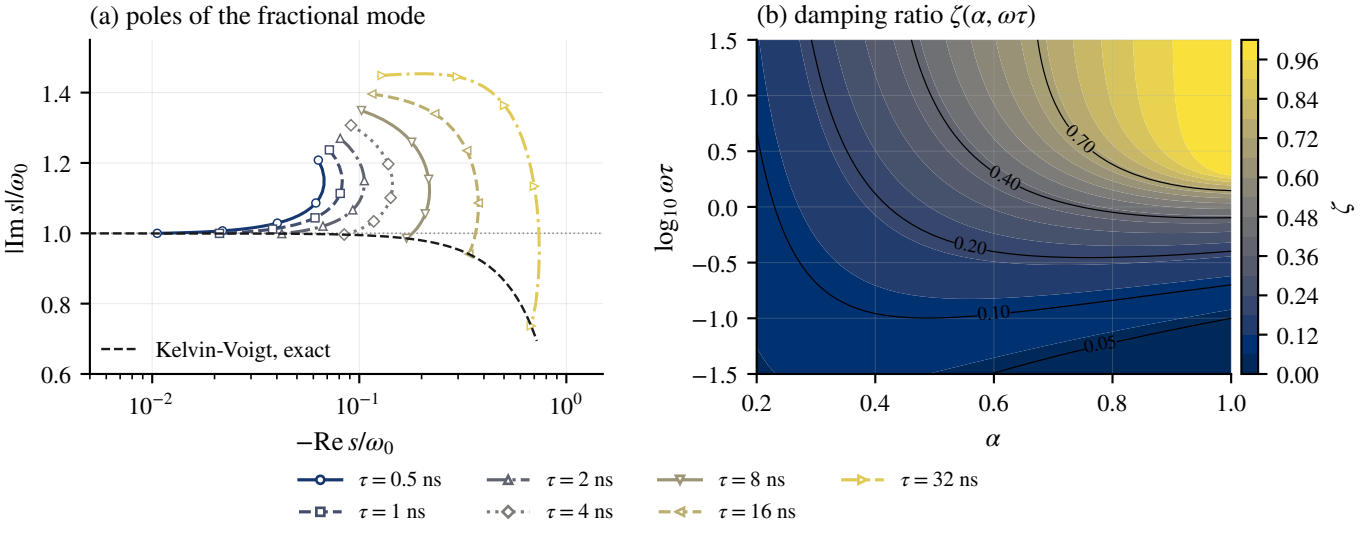
(a) poles of the fractional mode
|Im $s$|/$\omega_0$
1.4
1.2
1.0
0.8
0.6
Kelvin-Voigt, exact
$10^{-2}$
$10^{-1}$
$10^{0}$
−Re $s/\omega_0$
(b) damping ratio $\zeta(\alpha, \omega\tau)$
$\log_{10} \omega\tau$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.70
0.40
0.20
0.10
0.05
0.2
0.4
0.6
0.8
1.0
$\alpha$
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
$\zeta$
$\tau = 0.5$ ns
$\tau = 1$ ns
$\tau = 2$ ns
$\tau = 4$ ns
$\tau = 8$ ns
$\tau = 16$ ns
$\tau = 32$ ns